An algorithmic-level production-centered account of whole-word durations

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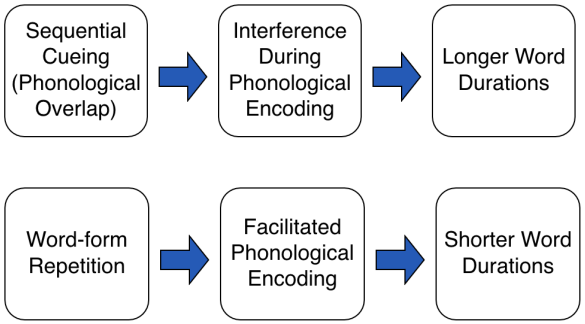
Introduction

- Phonological encoding is a complex, multi-step process that is sensitive to context
- Word durations can be a useful window into the dynamics of phonological encoding
- Major set of findings in phonological sequencing need computational explanations
  - Sequential Cueing
    - Production of words that have onset-position phonological overlap with recently produced or heard words results in slowed speech rate and longer word durations [1, 2]
  - Repetition Reduction
    - Repetition of word-forms that have recently been produced or heard by the talker results in shorter word durations [3]

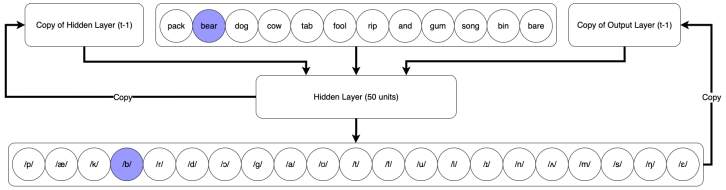
The Incremental Sequence Learning and Activation (ISLA) Theory

Hypothesis: Repetition reduction and the sequential cueing effect reflect incremental learning following the processing of phonological sequences



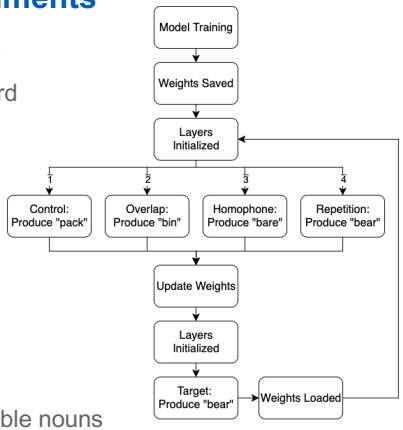
Modeling Segmental Spellout

- We implement a computational model to verify that the predictions of this hypothesis match human productions' word duration patterns
  - Simple Recurrent Network [4]
  - Error is a proxy for production difficulty
- Model *learns and updates weights* after every time it processes a phonological sequence



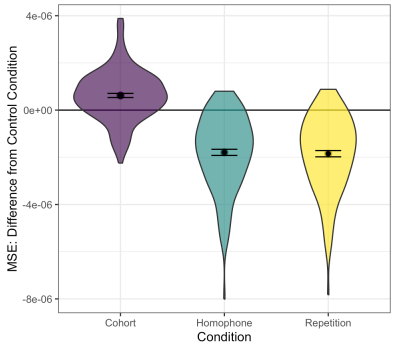
Experiments

- Train model to criterion
- Learns to produce sequence of segments for a given target word
- Run models through "lab experiment"

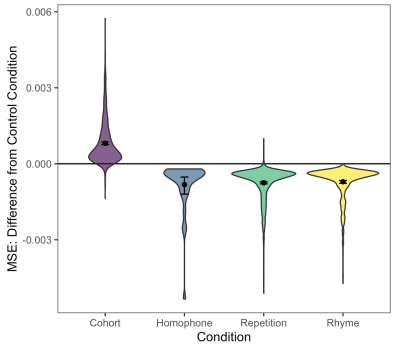


- **Experiment 1**
  Toy vocabulary
  + Control
  + Sequential cueing
  + Homophone
  + Repetition reduction
- **Experiment 2**
  1000 most frequent single-syllable nouns
  + Rhyme condition

Experiment 1 – Results



Experiment 2 – Results



Conclusions

- Incremental sequence learning model captures fundamental findings for overlap-driven lengthening and repetition reduction
- Successfully reproduces previous experimental patterns
- Lexical-level lingering activation or competition is not necessary to produce these effects

References

[1] Sevald, C. A., & Dell, G. S. (1994). The sequential cuing effect in speech production. *Cognition*, *53*(2), 91-127. [2] Myers, B. R., Jacobs, C. L., Buxó-Lugo, A., & Watson, D. G. (2025). Tomato-tomahto: Phonological representations vs. surface-level features in speech planning. *Journal of Memory and Language*, *143*, 104649. [3] Bard, E. G., Anderson, A. H., Sotillo, C., Aylett, M., Doherty-Sneddon, G., & Newlands, A. (2000). Controlling the intelligibility of referring expressions in dialogue. *Journal of memory and language*, *42*(1), 1-22. [4] Elman, J. L. (1990). Finding structure in time. *Cognitive science*, *14*(2), 179-211.